

# GVPT622 Lab Session

## Research Design

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- ▶ We'll reuse the Center for Effective Lawmaking (CEL) House dataset from last week to illustrate how randomization balances groups

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- ▶ Two main approaches:
  - ▷ **Experiments**: the researcher controls who receives a treatment
  - ▷ **Observational studies**: the researcher observes treatment assignment that occurred without their control
- ▶ Experiments are the considered the best designs to established causal claims, however observational studies are far more common practice

## Experiments

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## Case Study: Does Money Buy Access to Members of Congress?

- ▶ Kalla and Broockman (2016) ask: do campaign contributions help donors get meetings with congressional officials?
- ▶ A political advocacy organization asked its donor-members to request meetings with their own representatives' offices
- ▶ 191 congressional offices received a meeting request; the researchers randomly assigned whether the email revealed that the attendees were campaign donors

Revealing donor status → More access to senior congressional staff

- ▶ We cannot observe the counterfactual: would the same office have granted a meeting to the same group of constituents if they *hadn't* mentioned they were described donors?

# The Fundamental Problem in Causal Inference

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# The Fundamental Problem in Causal Inference

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- ▶ We only ever see one version of events for each office — the one that actually happened
- ▶ Because of this, establishing the causal effect of donor status is difficult

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  - ▷ Office  $X$ , request labeled "local constituent"  $\rightarrow$  meeting? (Yes/No)
- ▶ In practice, each office only ever receives *one* version of the request
- ▶ The factual outcome is observed; the counterfactual outcome stays hidden

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- ▶ Random assignment makes the two groups of offices comparable, on average, before treatment — similar seniority, ideology, district partisanship, etc
- ▶ Any systematic difference in meeting outcomes can then be attributed to donor status, not preexisting differences between offices (senior offices make themselves more available than junior offices, etc)

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- ▶ Randomly split all House seats held during the 118th Congress into two groups:

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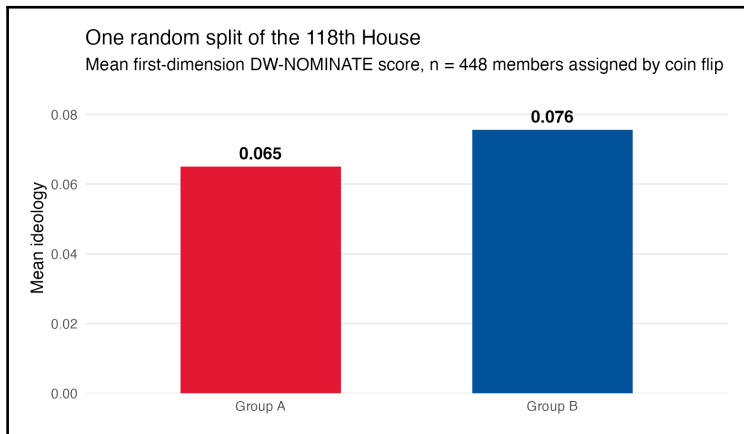
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# Illustrating Randomization

- ▶ We don't have Kalla and Broockman's raw office-level data, but we can illustrate the same logic with the CEL House dataset
- ▶ Randomly split all House seats held during the 118th Congress into two groups:

```
cel <- rename(cel,  
              congress = "Congress number",  
              ideology = "First-dimension DW-NOMINATE score")  
house118 <- filter(cel, congress == 118, !is.na(ideology))  
house118$group <- sample(rep(c("Group A", "Group B"),  
                             length.out = nrow(house118)))  
  
house118 |>  
  group_by(group) |>  
  summarise(mean_ideology = mean(ideology))
```

# Illustrating Randomization



A single random split of real 118th Congress House seats ( $n = 448$ ): average ideology is 0.065 in Group A and 0.076 in Group B, nearly identical, even though group membership was assigned by a coin flip.

## One Split, or a Thousand?

- ▶ One balanced split could be luck, so wrap it in a function and run it 1,000 times:

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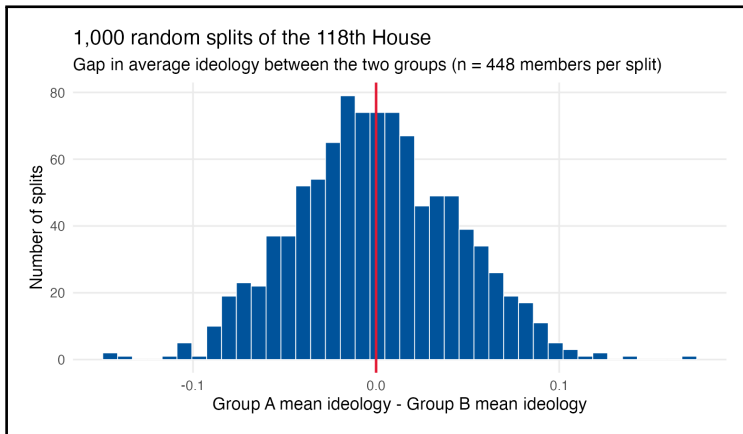
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# One Split, or a Thousand?

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```
random_split <- function(trial) {  
  house118 |>  
    mutate(group = sample(rep(c("A", "B"),  
                             length.out = n())) |>  
      group_by(group) |>  
      summarise(mean_ideology = mean(ideology)) |>  
      mutate(trial_no = trial)  
}  
  
splits <- map(1:1000, random_split) |> bind_rows()  
  
splits |>  
  group_by(group) |>  
  summarise(mean_mean_ideology = mean(mean_ideology))
```

# One Split, or a Thousand?



Average ideology across 1,000 random splits: 0.070 in Group A, 0.070 in Group B. Any one split can land off-center, but the gaps pile up on zero, randomization doesn't guarantee balance in a given trial, it guarantees it on average across repeated randomizations.

- ▶ Simulations are a great way to illustrate research designs with fake data.

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- ▶ The following exercise uses **FAKE** data

## Simulating Both Parallel Worlds

- ▶ Give all 10,000 simulated meeting requests *both* outcomes — the meeting they'd get as a revealed donor, and the one they'd get as a plain constituent:

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```
n <- 10000

sim_access <- tibble(
  in_state = rbinom(n, 1, 0.35)
) |>
  mutate(
    meeting_donor      = rbinom(n(), 1, if_else(in_state == 1, 0.35,
    0.55)),
    meeting_constituent = rbinom(n(), 1, if_else(in_state == 1, 0.45,
    0.20))
  )
```

- ▶ 35% of requests come from organizations based in the member's home state, and home-state status changes *both* potential outcomes

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- ▶ Because we wrote the probabilities down ourselves, we can do the impossible thing: read the true effect straight off the design
  - ▷ In-state (35%): 35% as donors vs. 45% as constituents
  - ▷ Out-of-state (65%): 55% vs. 20%
  - ▷ Average:  $0.35(-10) + 0.65(+35) = +19.25$  points

# What a Researcher Actually Observes

Each request only ever lives in one world

Did the office grant a meeting? The counterfactual cell stays hidden

|                |            |    |     |     |    |    |     |     |    |     |
|----------------|------------|----|-----|-----|----|----|-----|-----|----|-----|
| Constituent    | ?          | ?  | ?   | ?   | No | No | Yes | Yes | ?  | ?   |
| Revealed donor | No         | No | Yes | Yes | ?  | ?  | ?   | ?   | No | Yes |
|                | 1          | 2  | 3   | 4   | 5  | 6  | 7   | 8   | 9  | 10  |
|                | Request ID |    |     |     |    |    |     |     |    |     |

Simulated potential outcomes for the first 10 requests. Each column has exactly one cell we can see and one we can't — and which one is hidden was decided by a coin flip, not by anything about the office. That coin flip is the `revealed_donor` line in `run_experiment()`, on the next code slide.

- ▶ **Difference-of-means test:** compare the average outcome in the treatment group to determine if an intervention causes a statistically significant effect

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- ▶ **Difference-of-means test**: compare the average outcome in the treatment group to determine if an intervention causes a statistically significant effect
- ▶ In our example, we can compare the share of *Revealed Donor* offices that granted a meeting to the share of *Constituent* offices that granted a meeting
- ▶ Because random assignment balances the groups on everything else, this difference estimates the **average treatment effect** of revealing donor status

## Does the Difference of Means Find the Truth?

- ▶ Flip a coin for each request, 1,000 times:

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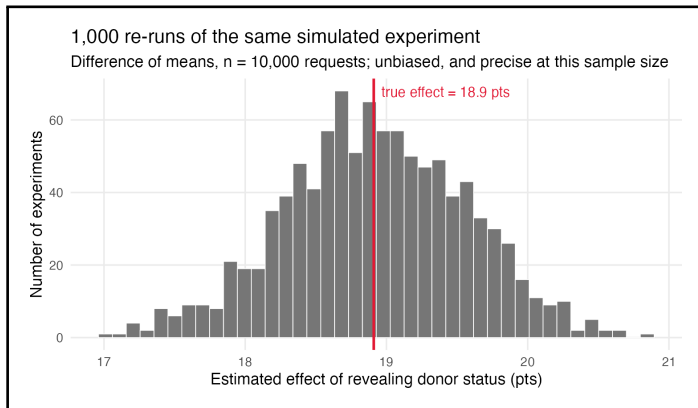
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# Does the Difference of Means Find the Truth?

- Flip a coin for each request, 1,000 times:

```
run_experiment <- function(trial) {  
  sim_access |>  
    mutate(revealed_donor = rbinom(n(), 1, 0.5),  
           observed = if_else(revealed_donor == 1,  
                              meeting_donor,  
                              meeting_constituent)) |>  
  group_by(revealed_donor) |>  
  summarise(rate = mean(observed), .groups = "drop") |>  
  summarise(estimate = diff(rate)) |>  
  mutate(trial_no = trial)  
}  
  
estimates <- map(1:1000, run_experiment) |> bind_rows()  
  
mean(estimates$estimate)
```

# Does the Difference of Means Find the Truth?



Across 1,000 experiments the estimates average  $\sim 18.9$  points. Randomization buys you the right answer on average.

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- ▶ **Constituent** condition: 2.4% of requests got a meeting with the Member or chief of staff
- ▶ **Revealed Donor** condition: 12.5% of requests did, a 10.1 percentage point increase
- ▶ Broadly speaking, senior policymakers made themselves available roughly **3–4 times** more often when individuals identified themselves as donors

## Observational Studies

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- ▶ Depending on the research design, observational studies can support causal claims
- ▶ Kalla and Broockman's design was experimental because they controlled which offices learned about donor status
- ▶ Lots of research cannot randomize the treatment; we usually observe what happened

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- ▶ Confounding variables can influence both whether a request mentions donor status and whether it succeeds
  - ▷ A study examining whether an individual's level of education (IV) is explanatory of their participation in an election (DV)
  - ▷ Income or Socioeconomic status would be considered an outside factor influencing both the IV and DV

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- ▶ Out-of-state organizations, who start from a lower baseline of access, may lean more heavily on flagging their donor status to get attention
- ▶ Comparing “mentioned donor status” to “didn't mention it” without accounting for this would give a misleading picture of donor status's effect

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- ▶ Observational analyses require controlling for variables that affect both treatment and outcome
- ▶ In our example, whether the requesting organization is headquartered in the member's home state
- ▶ When we control for home-state status, the true relationship between donor status and access becomes clearer

## Empirical Example (Simulated Data)

- ▶ *Not* from Kalla and Broockman — a simulated dataset built so that mentioning donor status has the same true effect (+15 points) for in-state and out-of-state organizations, to show how pooling can mask it:

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```
n <- 10000

sim <- tibble(
  in_state = rbinom(n, 1, 0.35),
  mention_prob = if_else(in_state == 1, 0.25, 0.55)
) |>
mutate(
  donor_mention = rbinom(n, 1, mention_prob),
  baseline_prob = if_else(in_state == 1, 0.35, 0.12),
  accept_prob    = baseline_prob + 0.15 * donor_mention,
  accepted       = rbinom(n, 1, accept_prob)
)
```

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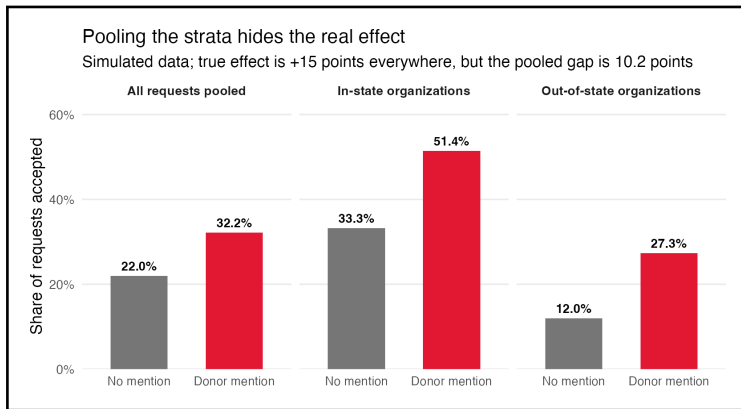
## Pooled vs. Stratified Comparison

- ▶ Note that here, unlike in the experiment, `mention_prob` depends on `in_state`
- ▶ Assignment is not random, so the comparison is not the treatment effect:

```
sim |>
  group_by(donor_mention) |>
  summarise(rate = mean(accepted), .groups = "drop")

sim |>
  group_by(in_state, donor_mention) |>
  summarise(rate = mean(accepted), .groups = "drop") |>
  group_by(in_state) |>
  mutate(difference = scales::percent(rate - lag(rate)))
```

# Empirical Example (Simulated Data)



Unadjusted difference: 10.2 points. Stratified by home-state status: 18.1 points (in-state) and 15.3 points (out-of-state) — both far closer to the true simulated effect of 15 points than the pooled number is. Pooling them together masks the real effect.

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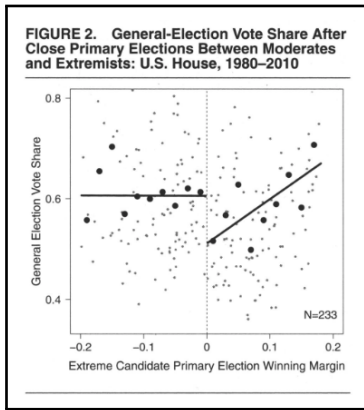
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  - ▷ Regression Discontinuity Designs
  - ▷ Difference-in-Differences
  - ▷ Matching
  - ▷ Instrumental Variables

- ▶ Quasi-experimental research design that estimates the causal effect of an intervention by comparing outcomes just above and just below a specific cutoff threshold

- ▶ Hall, Andrew B. 2015. "What Happens When Extremists Win Primaries?" *American Political Science Review*, 109(1), 18-42.

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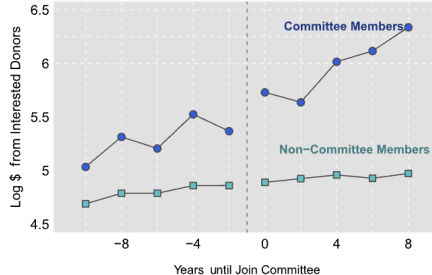
- ▶ Quasi-experimental research design that estimates the causal effect of an intervention by comparing the change in outcomes over time for a treated group to the change in outcomes over time for an untreated control group

- ▶ Fourinaies, Alexander, and Andrew B. Hall. 2018. “How Do Interest Groups Seek Access to Committees?” *American Journal of Political Science*, 62(1), 132-147.

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**FIGURE 1** Effect of Committee Membership on Contributions from Interested Donors

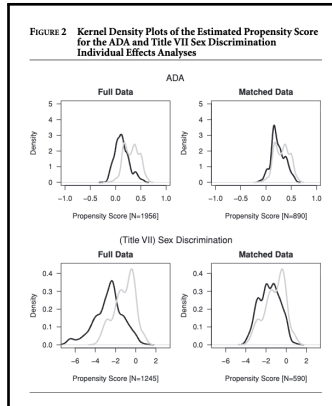


- ▶ Pairs treated and untreated units with similar observed characteristics to mimic a randomized experiment

- ▶ Boyd, Christina L., Lee Epstein, and Andrew D. Martin. 2010. "Untangling the Causal Effects of Sex on Judging." *American Journal of Political Science*, 54(2), 389-411.

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- ▶ Quasi-experimental design that uses an external, as-if random factor (the instrument) to isolate unbiased causal effects from observational data when direct treatment randomization is impossible

- ▶ Erikson, Robert S., and Laura Stoker. 2011. "Caught in the Draft: The Effects of Vietnam Draft Lottery Status on Political Attitudes." *American Political Science Review*, 105(2), 221-237.

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American Political Science Review

Vol. 105, No. 2 May 2011

doi:10.1017/S0003055411000141

## Caught in the Draft: The Effects of Vietnam Draft Lottery Status on Political Attitudes

ROBERT S. ERIKSON *Columbia University*

LAURA STOKER *University of California at Berkeley*

**T**he 1969 Vietnam draft lottery assigned numbers to birth dates in order to determine which young men would be called to fight in Vietnam. We exploit this natural experiment to examine how draft vulnerability influenced political attitudes. Data are from the Political Socialization Panel Study, which surveyed high school seniors from the class of 1965 before and after the national draft lottery was instituted. Males holding low lottery numbers became more antiwar, more liberal, and more Democratic in their voting compared to those whose high numbers protected them from the draft. They were also more likely than those with safe numbers to abandon the party identification that they had held as teenagers. Trace effects are found in reinterviews from the 1990s. Draft number effects exceed those for preadult party identification and are not mediated by military service. The results show how profoundly political attitudes can be transformed when public policies directly affect citizens' lives.

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- ▶ The goal is to show that changes in one variable (IV) cause changes in an outcome of interest (DV)
- ▶ We need groups that are identical except for treatment assignment
- ▶ Field experiments, like Kalla and Broockman's, achieve this through random assignment
- ▶ Observational studies require careful statistical control to approximate comparable groups and detailed research design to identify causal relationships

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**Questions?**